MSG No. 52.315 P_bnna [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,2,11,12]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,4,9,10]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[5,6,15,16]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,8,13,14]`

Table 2: Wyckoff site: `4b`, site symmetry: `22'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,4,9,10]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[5,6,15,16]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,8,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,9,12]`
3	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[6,7,13,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[1,4,10,11]`
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[6,7,13,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,13]`
2	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[2,14]`
3	$\left[0, \frac{3}{4}, 0\right]$	`[3,15]`
4	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[4,16]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,9]`
6	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,10]`
7	$\left[0, \frac{1}{4}, 0\right]$	`[7,11]`

8	$\left[0,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[8,12]`

Table 6: Wyckoff site: `8f`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,6]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,7]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,8]`
5	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,13]`
6	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[10,14]`
7	$[0,\ 0,\ 0]$	`[11,15]`
8	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[12,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[3,4]`
3	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[5,6]`
4	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[7,8]`
5	$\left[x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[9,10]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[11,12]`
7	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[13,14]`
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[15,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[1,10]`
2	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[4,11]`
5	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[6,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[8,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,12]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[2,11]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[3,10]`

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	[4,9]
5	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	[5,16]
6	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[6,15]
7	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[7,14]
8	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	[8,13]

Table 10: Wyckoff site: 8j, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{3}{4}, 0, z\right]$	[1,4]
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}-z\right]$	[2,3]
3	$\left[\frac{1}{4}, 0, -z\right]$	[5,8]
4	$\left[\frac{1}{4}, \frac{1}{2}, z+\frac{1}{2}\right]$	[6,7]
5	$\left[\frac{3}{4}, \frac{1}{2}, z\right]$	[9,12]
6	$\left[\frac{3}{4}, 0, \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{1}{4}, \frac{1}{2}, -z\right]$	[13,16]
8	$\left[\frac{1}{4}, 0, z+\frac{1}{2}\right]$	[14,15]

Table 11: Wyckoff site: 8k, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{3}{4}, y, \frac{3}{4}\right]$	[1,11]
2	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	[2,12]
3	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{3}{4}\right]$	[3,9]
4	$\left[\frac{3}{4}, -y, \frac{3}{4}\right]$	[4,10]
5	$\left[\frac{1}{4}, -y, \frac{1}{4}\right]$	[5,15]
6	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[6,16]
7	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{4}\right]$	[7,13]
8	$\left[\frac{1}{4}, y, \frac{1}{4}\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{3}{4}, y, \frac{1}{4}\right]$	[1,11]
2	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{4}\right]$	[2,12]
3	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{4}\right]$	[3,9]
4	$\left[\frac{3}{4}, -y, \frac{1}{4}\right]$	[4,10]
5	$\left[\frac{1}{4}, -y, \frac{3}{4}\right]$	[5,15]
6	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{3}{4}\right]$	[6,16]
7	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{3}{4}\right]$	[7,13]
8	$\left[\frac{1}{4}, y, \frac{3}{4}\right]$	[8,14]

Table 13: Wyckoff site: `16m`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[6]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x+\frac{1}{2},\ y,\ -z\right]$	`[8]`
9	$\left[x,\ y+\frac{1}{2},\ z\right]$	`[9]`
10	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	`[10]`
11	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[12]`
13	$\left[-x,\ \frac{1}{2}-y,\ -z\right]$	`[13]`
14	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[14]`
15	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[16]`